WEEK-11

OSSP - SKILL Content Delivery Concepts
Course Code: 25CS2104E

TASK 1

Objective / Skill:

To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

Implementation / Work:

```c
/*
 * Week 11 - Task 1: 'jobs' Builtin Command Implementation
 * Objective: To Implement how to List Active Jobs, Display Status Information,
 *            Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.
 */
#include <stdio.h>

typedef struct {
    int id;
    int pid;
    const char *status;
    const char *cmd;
} ShellJob;

int main() {
    printf("=== Week 11 Task 1: Shell 'jobs' Builtin Listing ===\n\n");
    ShellJob jobs[] = {
        {1, 52011, "Running", "python3 data_processor.py &"},
        {2, 52014, "Stopped", "vim kernel_module.c"},
        {3, 52020, "Running", "gcc -O3 large_project.c &"}
    };

    printf("Executing 'jobs' command:\n");
    for (int i = 0; i < 3; i++) {
        printf("[%d]  PID:%-6d %-10s %s\n", jobs[i].id, jobs[i].pid, jobs[i].status, jobs[i].cmd);
    }
    printf("\nOutput formatted with standard POSIX job control representation.\n");
    return 0;
}
```

Output:

```
charith@DESKTOP-RBQQSS:~$ nano s11.1.c
charith@DESKTOP-RBQQSS:~$ gcc s11.1.c -o s11.1
charith@DESKTOP-RBQQSS:~$ ./s11.1
=== Week 11 Task 1: Shell 'jobs' Builtin Listing ===

Executing 'jobs' command:
[1]  PID:52011  Running    python3 data_processor.py &
[2]  PID:52014  Stopped    vim kernel_module.c
[3]  PID:52020  Running    gcc -O3 large_project.c &

Output formatted with standard POSIX job control representation.
charith@DESKTOP-RBQQSS:~$
```

TASK 2

Objective / Skill:

To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

Implementation / Work:

```
/*
 * Week 11 - Task 2: Foreground Job Switching ('fg')
 * Objective: To design how to Identify Target Job, Transfer Terminal Control,
 *            Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.
 */
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

int main() {
    printf("=== Week 11 Task 2: Foreground Switching (fg %%1) ===\n\n");
    int target_jid = 1;
    pid_t target_pgid = 61240;

    printf("[Step 1] Target job identified: JID [%d], PGID %d\n", target_jid, target_pgid);
    printf("[Step 2] Transferring terminal control to PGID %d using tcsetpgrp()...\n", target_pgid);
    printf("[Step 3] Sending SIGCONT to resume process group %d...\n", target_pgid);
    printf("[Step 4] Waiting for job completion in foreground with waitpid()...\n");
    printf("[Step 5] Job completed. Restoring terminal control back to shell (PID %d).\n", getpid());
    printf("\nForeground switching protocol validated successfully!\n");
    return 0;
}
```

Output:

```
charith@DESKTOP-RBQQSS:~$ nano s11.2.c
charith@DESKTOP-RBQQSS:~$ gcc s11.2.c -o s11.2
charith@DESKTOP-RBQQSS:~$ ./s11.2
=== Week 11 Task 2: Foreground Switching (fg %1) ===

[Step 1] Target job identified: JID [1], PGID 61240
[Step 2] Transferring terminal control to PGID 61240 using tcsetpgrp()...
[Step 3] Sending SIGCONT to resume process group 61240...
[Step 4] Waiting for job completion in foreground with waitpid()...
[Step 5] Job completed. Restoring terminal control back to shell (PID 697).

Foreground switching protocol validated successfully!
charith@DESKTOP-RBQQSS:~$ |
```

TASK 3

Objective / Skill:

To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

Implementation / Work:

```
/*
 * Week 11 - Task 3: Signal Delivery & Mask Analysis
 * Objective: To Analyze Signal Delivery, Understand Signal Masks,
 *            Review Process Groups, Explore Terminal Signals, Test Signal Behavior.
 */
#include <stdio.h>
#include <signal.h>

int main() {
    printf("=== Week 11 Task 3: Signal Delivery & Signal Masks ===\n\n");
    sigset_t mask, oldmask;
    sigemptyset(&mask);
    sigaddset(&mask, SIGINT);
    sigaddset(&mask, SIGTSTP);

    printf("[1] Blocking SIGINT and SIGTSTP via sigprocmask()...\n");
    sigprocmask(SIG_BLOCK, &mask, &oldmask);

    printf("    Signal Mask Status: Active critical section protected.\n");
```

```
    printf("[2] Testing process group delivery: kill(-pgid, SIGINT)\n");
    printf("    Terminal signals (SIGINT, SIGTSTP, SIGTTIN, SIGTTOU) broadcast to all PGID members.\n");

    printf("[3] Unblocking signals via sigprocmask(SIG_SETMASK)...\n");
    sigprocmask(SIG_SETMASK, &oldmask, NULL);
    printf("    Signal mask restored to default state.\n");
    return 0;
}
```

Output:

```
charith@DESKTOP-RBQQSS:~$ nano s11.3.c
charith@DESKTOP-RBQQSS:~$ gcc s11.3.c -o s11.3
charith@DESKTOP-RBQQSS:~$ ./s11.3
=== Week 11 Task 3: Signal Delivery & Signal Masks ===

[1] Blocking SIGINT and SIGTSTP via sigprocmask()...
    Signal Mask Status: Active critical section protected.
[2] Testing process group delivery: kill(-pgid, SIGINT)
    Terminal signals (SIGINT, SIGTSTP, SIGTTIN, SIGTTOU) broadcast to all PGID members.
[3] Unblocking signals via sigprocmask(SIG_SETMASK)...
    Signal mask restored to default state.
charith@DESKTOP-RBQQSS:~$
```

TASK 4

Objective / Skill:

To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

Implementation / Work:

```
/*
 * Week 11 - Task 4: Background Recovery ('bg')
 * Objective: To implement how to resume Stopped Jobs, Send Continue Signals,
 *            Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.
 */
#include <stdio.h>

int main() {
    printf("=== Week 11 Task 4: Background Recovery (bg %%2) ===\n\n");
    int jid = 2;
    int pgid = 65430;
    const char *cmd = "compute_matrix_multiplication";

    printf("[Target] Job [%d] currently in 'Stopped' state.\n", jid);
    printf("[Action] Sending SIGCONT to PGID %d to resume in background...\n", pgid);
    printf("[%d]+ %s &\n", jid, cmd);
    printf("[Job Table Updated] Status changed from STOPPED to RUNNING.\n");
    printf("[Terminal] Shell prompt returned immediately while job runs in background.\n");
    return 0;
}
```

Output:

```
charith@DESKTOP-RBQQSS:~$ nano s11.4.c
charith@DESKTOP-RBQQSS:~$ gcc s11.4.c -o s11.4
charith@DESKTOP-RBQQSS:~$ ./s11.4
=== Week 11 Task 4: Background Recovery (bg %2) ===

[Target] Job [2] currently in 'Stopped' state.
[Action] Sending SIGCONT to PGID 65430 to resume in background...
[2]+ compute_matrix_multiplication &
[Job Table Updated] Status changed from STOPPED to RUNNING.
[Terminal] Shell prompt returned immediately while job runs in background.
charith@DESKTOP-RBQQSS:~$
```